

PROBLEM SET 1—SOLUTIONS

ECON301—ECONOMETRICS I

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These are my suggested solutions. It is your responsibility to cross-check this document with your own answers.

1. (a) By the law of total probability we have that $f(x, y) = f(y|x)f_X(x)$. I am going to keep things relatively general regarding the computation of the joint distribution, i.e., I am going to keep λ as a variable and only until the end I will substitute the value of $\lambda = 0.7$. This is so you can see how to work with these kind of integrals in a very efficient way. Since¹ $f_X(x) = \frac{\lambda^2 x e^{-\lambda x}}{\Gamma(2)} 1_{[0, \infty)}(x) = \lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x)$, we have that

$$f(x, y) = \left(\frac{1}{x} 1_{[0, x]}(y)\right) \left(\lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x)\right) = \lambda^2 e^{-\lambda x} 1_{[0, x]}(y) 1_{[0, \infty)}(x)$$

- (b) We need to integrate over x the joint distribution:

$$f_Y(y) = \int_{-\infty}^{\infty} \lambda^2 e^{-\lambda x} 1_{[0, x]}(y) 1_{[0, \infty)}(x) dx$$

Note that in the above integral, both indicator functions depend on x : $1_{[0, \infty)}(x)$ through its argument, and $1_{[0, x]}(y)$ through the set the indicator is referring to. Thus, we cannot proceed with the integral as it is. Note that the joint pdf is defined over the region in which x varies from 0 to infinity and y varies from 0 to x . This corresponds to the area described by the lower triangle in the positive plane bounded by the axes and the line $y = x$. The product of the indicators functions $1_{[0, x]}(y) 1_{[0, \infty)}(x)$ describes this region. We can alternatively describe this region as y varying from 0 to infinity and x varying from y to infinity. Thus, we can alternatively write the indicator functions as $1_{[0, \infty)}(y) 1_{[y, \infty)}(x)$. In this way, the integration variable x appears only as an argument and not in the description of the set for an indicator function. Thus, we can proceed with the integration of the expression above:

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} \lambda^2 e^{-\lambda x} 1_{[0, x]}(y) 1_{[0, \infty)}(x) dx = \int_{-\infty}^{\infty} \lambda^2 e^{-\lambda x} 1_{[0, \infty)}(y) 1_{[y, \infty)}(x) dx = \int_y^{\infty} \lambda^2 e^{-\lambda x} 1_{[0, \infty)}(y) dx \\ &= 1_{[0, \infty)}(y) \lambda^2 \left(-\frac{1}{\lambda} e^{-\lambda x}\right)_y^{\infty} = -1_{[0, \infty)}(y) \lambda \left(\lim_{x \rightarrow \infty} e^{-\lambda x} - e^{-\lambda y}\right) = -1_{[0, \infty)}(y) \lambda (0 - e^{-\lambda y}) \\ &= \lambda e^{-\lambda y} 1_{[0, \infty)}(y) \end{aligned}$$

As you can see, the marginal distribution of Y is an exponential distribution with parameter λ .

- (c) We need to integrate over y the joint distribution and since the indicator functions include y only as an argument, we can proceed easily:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} \lambda^2 e^{-\lambda x} 1_{[0, x]}(y) 1_{[0, \infty)}(x) dy = \int_0^x \lambda^2 e^{-\lambda x} 1_{[0, \infty)}(x) dy = \lambda^2 e^{-\lambda x} 1_{[0, \infty)}(x) (y)_0^x \\ &= \lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x) \end{aligned}$$

As expected, we obtained the same Gamma distribution as the one given initially.

- (d) We will let $w(x) = E[Y|X = x]$. To compute the conditional expectation, we will use the conditional distribution:

$$w(x) = E[Y|X = x] = \int_{-\infty}^{\infty} \frac{y}{x} 1_{[0, x]}(x) dy = \int_0^x \frac{y}{x} dy = \left(\frac{y^2}{2x}\right)_0^x = \frac{x}{2}$$

Note that the conditional expectation $w(x)$ is linear.

- i. $w(2.5) = 1.25$, $w(5) = 2.5$, $w(10) = 5$. Comparing these results with the simulation in R, you can see that the simulation values are quite close to the theoretical values here.
- (e) We will verify the law of the iterated expectations for this example.

¹Recall that the Gamma function Γ extends the notion of the factorial to all complex numbers. In particular, for all natural numbers $n \in \mathbb{N}$, we have that $\Gamma(n) = (n-1)!$.

- i. Using the marginal distribution $f_Y(y)$ we get

$$E(Y) = \int_{-\infty}^{\infty} y \lambda e^{-\lambda y} 1_{[0, \infty)}(y) dy = \int_0^{\infty} y \lambda e^{-\lambda y} dy$$

We could continue with the integral by solving it by integration by parts. However, I am going to show you how to use a common trick in probability to quickly obtain these type of integrals. Note that the integrand looks almost like the integrand of a Gamma distribution but it is missing some constants that would make it a true distribution. What we are going to do is to multiply the expression by a carefully constructed factor that will equal 1 but that will convert the integrand into a true Gamma distribution. For the above expression to be a Gamma distribution with parameters 2 and λ , it needs to have an exponent of 2 in the λ parameter. Thus, we are going to multiply and divide the expression by λ :

$$= \frac{\lambda}{\lambda} \int_0^{\infty} y \lambda e^{-\lambda y} dy = \frac{1}{\lambda} \int_0^{\infty} y \lambda^2 e^{-\lambda y} dy$$

The expression inside the integral, being a distribution function, integrates to 1 when integrated over the whole support of the distribution, thus we get:

$$= \frac{1}{\lambda}$$

When $\lambda = 0.7$, then $E(Y) = 1/0.7 = 1.4286$.

- ii. Recall that $w(x) = E(Y|X = x)$. If we consider the argument as random, then $w(X)$ is a random variable of X . Thus, when computing $E[w(X)]$, we will use the marginal distribution of X :

$$E[w(X)] = E[E(Y|X)] = \int_{-\infty}^{\infty} (x/2) \lambda^2 x e^{-\lambda x} 1_{[0, \infty)} dx = \int_0^{\infty} \frac{\lambda^2}{2} x^2 e^{-\lambda x} dx$$

Note that the integrand is almost like the integrand from a Gamma distribution with parameters 3 and λ , but is missing an exponent of 3 in λ . Thus, using a similar trick as above

$$= \frac{\lambda}{\lambda} \int_0^{\infty} \frac{\lambda^2}{2} x^2 e^{-\lambda x} dx = \frac{1}{\lambda} \int_0^{\infty} \frac{\lambda^3}{2} x^2 e^{-\lambda x} dx$$

and since the integrand now integrates to 1 over the support of the distribution we have that

$$= \frac{1}{\lambda}$$

When $\lambda = 0.7$ we have that $E[w(X)] = 1/0.7 = 1.4286$. Since $E[w(X)] = E[E(Y|X)] = E[Y]$, we have verified for this particular example the law of iterated expectations.

- (f) Recall that $\text{Var}(Y|X = x) = E[Y^2|X = x] - [E(Y|X = x)]^2$. We already have that $E[Y|X = x] = x/2$, we only need to compute $E[Y^2|X = x]$ using the conditional distribution.

$$E[Y^2|X = x] = \int_{-\infty}^{\infty} \frac{y^2}{x} 1_{[0, x]}(y) dy = \int_0^x \frac{y^2}{x} dy = \left(\frac{y^3}{3x} \right)_0^x = \frac{x^2}{3}.$$

Thus, $\text{Var}(Y|X = x) = \frac{x^2}{3} - \left(\frac{x}{2} \right)^2 = \frac{4x^2 - 3x^2}{12} = \frac{x^2}{12}$.

- (g) We will verify the law of iterated variances for this example.

- i. We will compute $\text{Var}(Y)$ by using the marginal distribution $f_Y(y)$. In order to compute the variance, we need $E[Y^2]$

$$\begin{aligned} E[Y^2] &= \int_{-\infty}^{\infty} y^2 \lambda e^{-\lambda y} 1_{[0, \infty)}(y) dy = \int_0^{\infty} y^2 \lambda e^{-\lambda y} dy = \frac{2\lambda^2}{2\lambda^2} = \int_0^{\infty} y^2 \lambda e^{-\lambda y} dy \\ &= \frac{2}{\lambda^2} \int_0^{\infty} \frac{\lambda^3}{2} y^2 e^{-\lambda y} dy = \frac{2}{\lambda^2}, \end{aligned}$$

where we have used the same trick as before, making the term in the last integrand equal to the pdf of a Gamma distribution with parameters 3 and λ , which therefore, integrates to 1. Therefore,

$$\text{Var}(Y) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda} \right)^2 = \frac{1}{\lambda^2}.$$

When $\lambda = 0.7$, $\text{Var}(Y) = 1/.7^2 = 2.0408$.

- ii. Let $w(x) = E[Y|X = x]$, and $h(x) = \text{Var}(Y|X = x)$, so that both $w(X)$ and $h(X)$ are random variables of X . We will compute $E[h(X)]$ and $\text{Var}(w(X))$. We start with $E[h(X)]$:

$$E[h(X)] = \int_{-\infty}^{\infty} \frac{x^2}{12} \lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x) dx = \int_0^{\infty} \frac{\lambda^2}{12} x^3 e^{-\lambda x} dx$$

Using the same trick as before, we will convert the integrand into the integrand of a Gamma distribution with parameters 4 and λ :

$$\begin{aligned} &= \int_0^{\infty} \frac{\lambda^2}{2(3!)} x^3 e^{-\lambda x} dx = \frac{\lambda^2}{2\lambda^2} \int_0^{\infty} \frac{\lambda^2}{(3!)} x^3 e^{-\lambda x} dx \\ &= \frac{1}{2\lambda^2} \int_0^{\infty} \frac{\lambda^4}{3!} x^3 e^{-\lambda x} dx = \frac{1}{2\lambda^2} \end{aligned}$$

To compute $\text{Var}(w(X)) = E[(w(X))^2] - (E[w(X)])^2$, we need to compute only the first term, since we have obtained $E[w(X)]$ in item (e.ii).

$$E[(w(X))^2] = E[(X/2)^2] = E[X^2/4] = \int_{-\infty}^{\infty} \frac{x^2}{4} \lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x) dx = \int_0^{\infty} \frac{\lambda^2}{4} x^3 e^{-\lambda x} dx$$

We will convert the integrand into the integrand of a Gamma distribution with parameters 4 and λ :

$$\begin{aligned} &= \frac{3!\lambda^2}{3!\lambda^2} \int_0^{\infty} \frac{\lambda^2}{4} x^3 e^{-\lambda x} dx = \frac{3!}{4\lambda^2} \int_0^{\infty} \frac{\lambda^4}{3!} x^3 e^{-\lambda x} dx \\ &= \frac{6}{4\lambda^2} = \frac{3}{2\lambda^2} \end{aligned}$$

Thus

$$\text{Var}(w(X)) = \frac{3}{2\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{3-2}{2\lambda^2} = \frac{1}{2\lambda^2}.$$

Therefore

$$E[h(X)] + \text{Var}(w(X)) = \frac{1}{2\lambda^2} + \frac{1}{2\lambda^2} = \frac{2}{2\lambda^2} = \frac{1}{\lambda^2},$$

which indeed is equal to $\text{Var}(Y)$ as computed directly from the marginal distribution $f_Y(y)$ in item (g.i).

- (h) We need to compute $\text{Cov}(X, Y)$, $E[X]$, and $\text{Var}(X)$. I start with $E[X]$:

$$\begin{aligned} E[X] &= \int_{-\infty}^{\infty} x \lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x) dx = \int_0^{\infty} \lambda^2 x^2 e^{-\lambda x} dx = \frac{2!\lambda}{2!\lambda} \int_0^{\infty} \lambda^2 x^2 e^{-\lambda x} dx \\ &= \frac{2!}{\lambda} \int_0^{\infty} \frac{\lambda^3}{2!} x^2 e^{-\lambda x} dx = \frac{2!}{\lambda} = \frac{2}{\lambda}. \end{aligned}$$

To compute $\text{Var}(X)$, we need to compute $E[X^2]$:

$$\begin{aligned} E[X^2] &= \int_{-\infty}^{\infty} x^2 \lambda^2 x e^{-\lambda x} 1_{[0, \infty)}(x) dx = \int_0^{\infty} \lambda^2 x^3 e^{-\lambda x} dx = \frac{3!\lambda^2}{3!\lambda^2} \int_0^{\infty} \lambda^2 x^3 e^{-\lambda x} dx \\ &= \frac{3!}{\lambda^2} \int_0^{\infty} \frac{\lambda^4}{3!} x^3 e^{-\lambda x} dx = \frac{6}{\lambda^2} \end{aligned}$$

Thus,

$$\text{Var}(X) = \frac{6}{\lambda^2} - \left(\frac{2}{\lambda}\right)^2 = \frac{6-4}{\lambda^2} = \frac{2}{\lambda^2}$$

To compute $\text{Cov}(X, Y)$, we need to compute $E[XY]$:

$$\begin{aligned}
E[XY] &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy\lambda^2 e^{-\lambda x} 1_{[0,x]}(y) 1_{[0,\infty)}(x) dy dx = \int_{-\infty}^{\infty} x\lambda^2 e^{-\lambda x} 1_{[0,\infty)}(x) \left(\int_0^x y dy \right) dx \\
&= \int_{-\infty}^{\infty} x\lambda^2 e^{-\lambda x} 1_{[0,\infty)}(x) (y^2/2)_0^x dx = \int_{-\infty}^{\infty} x\lambda^2 e^{-\lambda x} (x^2/2) 1_{[0,\infty)}(x) dx \\
&= \int_0^{\infty} \frac{\lambda^2}{2} x^3 e^{-\lambda x} dx = \frac{3\lambda^2}{3\lambda^2} \int_0^{\infty} \frac{\lambda^2}{2} x^3 e^{-\lambda x} dx \\
&= \frac{3}{\lambda^2} \int_0^{\infty} \frac{\lambda^4}{3 \cdot 2} x^3 e^{-\lambda x} dx = \frac{3}{\lambda^2} \int_0^{\infty} \frac{\lambda^4}{3!} x^3 e^{-\lambda x} dx \\
&= \frac{3}{\lambda^2}
\end{aligned}$$

Thus,

$$\text{Cov}(X, Y) = \frac{3}{\lambda^2} - \frac{2}{\lambda} \frac{1}{\lambda} = \frac{3-2}{\lambda^2} = \frac{1}{\lambda^2}$$

Therefore, we have that, for the BLP $E^*(Y|X) = \alpha + \beta X$, the parameters α and β are given by

$$\begin{aligned}
\beta &= \text{Cov}(X, Y) / \text{Var}(X) = \frac{\frac{1}{\lambda^2}}{\frac{2}{\lambda^2}} = \frac{1}{2}, \\
\alpha &= E(Y) - \beta E(X) = \frac{1}{\lambda} - \left(\frac{1}{2}\right) \left(\frac{2}{\lambda}\right) = \frac{1}{\lambda} - \frac{1}{\lambda} = 0.
\end{aligned}$$

Thus, the BLP is given by $E^*(Y|X) = X/2$.

- (i) It is clear from items (d) and (h) that the conditional expectation and the BLP are the same. This corroborates the result from theorem 2.9: since the conditional expectation is linear, the best linear predictor is equal to the conditional expectation.

2. Proof of Theorem 2.4. Using the properties (i), (iv) and (v) from theorem 2.3 we have that

$$\begin{aligned}
\text{Var}(X + Y|Z) &= \text{Cov}(X + Y, X + Y|Z) = \text{Cov}(X, X|Z) + \text{Cov}(X, Y|Z) + \text{Cov}(Y, X|Z) + \text{Cov}(Y, Y|Z) \\
&= \text{Var}(X|Z) + 2\text{Cov}(X, Y|Z) + \text{Var}(Y|Z).
\end{aligned}$$

3.

$$\begin{aligned}
E(X^m Y^n) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^m y^n (x + y) 1_{[0,1]}(x) 1_{[0,1]}(y) dy dx = \int_0^1 \int_0^1 x^{m+1} y^n dy dx + \int_0^1 \int_0^1 y^{n+1} x^m dy dx \\
&= \int_0^1 x^{m+1} \left(\frac{y^{n+1}}{n+1} \right)_0^1 dx + \int_0^1 x^m \left(\frac{y^{n+2}}{n+2} \right)_0^1 dx = \frac{1}{n+1} \int_0^1 x^{m+1} dx + \frac{1}{n+2} \int_0^1 x^m dx \\
&= \frac{1}{n+1} \left(\frac{x^{m+2}}{m+2} \right)_0^1 + \frac{1}{n+2} \left(\frac{x^{m+1}}{m+1} \right)_0^1 = \frac{1}{n+1} \frac{1}{m+2} + \frac{1}{n+2} \frac{1}{m+1} \\
&= \frac{(n+2)(m+1) + (n+1)(m+2)}{(n+1)(n+2)(m+1)(m+2)}
\end{aligned}$$

4.

$$\begin{aligned}
E(U^*|X_1, \dots, X_k) &= E[Y - E^*(Y|X_1, \dots, X_k)|X_1, \dots, X_k] \\
&= E(Y|X_1, \dots, X_k) - E[E^*(Y|X_1, \dots, X_k)|X_1, \dots, X_k] \\
&= E(Y|X_1, \dots, X_k) - E^*(Y|X_1, \dots, X_k)
\end{aligned}$$

where the last equality is because the BLP $E^*(Y|X_1, \dots, X_k)$ is only a function of the conditioning variables $X_1, \dots, X_k$. A simple example is the one given in lecture 2: consider the random vector (X, Y) with pdf $f(x, y) = (x + y) 1_{[0,1]}(x) 1_{[0,1]}(y)$. Then we have that $E[Y|X, X^2] = E[Y|X] = \frac{3x+2}{6x+3}$, since the conditional expectation will not change if we condition also on a function of X . However, the BLP when we use X and X^2 as variables is given by $E^*(Y|X, X^2) = 0.6568 - 0.19607 + 0.098039X^2$. It is obvious that the difference between these two is different from zero.

5. The BLP of Y as a function of X is given by $E^*(Y|X) = \alpha + \beta X$ where $\alpha = E(Y) - \beta E(X)$ and $\beta = \text{Cov}(X, Y)/\text{Var}(X)$. Now, let $Z = E(Y|X)$. The BLP of Z as a function of X is given by $E^*(Z|X) = \tilde{\alpha} + \tilde{\beta}X$, where

$$\begin{aligned}\tilde{\alpha} &= E(Z) - \tilde{\beta}E(X) \\ \tilde{\beta} &= \text{Cov}(Z, X)/\text{Var}(X)\end{aligned}$$

Note that $\text{Cov}(Z, X) = \text{Cov}(E(Y|X), X) = E[E(Y|X)X] - E[E(Y|X)]E[X] = E[E(YX|X)] - E[E(Y|X)]E[X] = E[YX] - E[Y]E[X] = \text{Cov}(Y, X)$; where the first equality is by definition of $Z = E(Y|X)$, the second equality is the computation identity for the covariance, the third equality is because when conditioning on X , it can be taken as a constant inside the conditional expectation, the fourth equality is by two applications of the law of iterated expectations, and finally the last equality is again by the computation identity of the covariance. The previous calculation implies that $\tilde{\beta} = \beta$.

Similarly, since $E(Z) = E(E(Y|X)) = E(Y)$ by the law of iterated expectations, we have that $\tilde{\alpha} = \alpha$. Therefore, we conclude that since the coefficients are the same, the BLPs in both cases coincide.

Intuitively, this means that the BLP cannot extract more information about the relationship between Y as explained by X than the one the conditional expectation is already extracting.